

VERESHCHAGINO PERM KRAI, Russian Federation

WMO: 282160

Lat: 58.083N

Lon: 54.683E

Elev: 743

StdP: 14.31

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-15.4	-26.2	1.3	-20.6	-20.5	1.8	-14.8	21.2	11.9	18.6	13.5	2.3	0	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	84.4	68.2	81.4	67.0	78.0	65.4	71.1	81.1	68.8	78.5	66.7	75.7	4.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	104.0	77.4	65.2	95.9	74.5	63.0	88.6	72.1	35.4	80.9	33.4	79.1	31.7	76.2	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.2	12.4	DB	-26.4	88.6	6.8	3.7	-31.3	91.2	-35.2	93.4	-39.0	95.4	-43.9	98.1	(4)
(5)			WB	-27.1	74.6	6.9	2.9	-32.1	76.7	-36.1	78.4	-40.0	80.0	-45.0	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	7.8	10.7	24.1	38.7	52.2	61.4	65.8	60.4	49.7	36.8	22.3	11.8
	DBStd	22.55	13.03	12.30	7.87	9.22	9.66	7.83	7.77	7.85	8.02	8.43	11.45	12.95
	HDD50	6214	1307	1101	803	356	94	8	2	7	100	419	832	1185
	HDD65	10457	1772	1521	1268	788	406	159	87	189	461	874	1282	1650
	CDD50	1452	0	0	0	17	163	350	492	328	92	10	0	0
	CDD65	222	0	0	0	0	10	51	112	46	3	0	0	0
	CDH74	1734	0	0	0	0	92	389	902	335	16	0	0	0
	CDH80	416	0	0	0	0	11	73	242	89	1	0	0	0
Wind	WSAvg	4.8	5.2	5.3	5.4	4.7	5.2	4.7	3.7	3.7	4.1	5.2	5.3	5.5
Precipitation	PrecAvg	23.3	1.6	1.0	1.5	1.3	1.7	2.6	2.7	2.6	2.3	2.1	2.0	1.7
	PrecMax	31.6	2.9	1.6	4.6	3.0	3.7	6.5	6.4	6.4	5.4	4.1	3.3	2.9
	PrecMin	16.0	0.3	0.4	0.1	0.0	0.5	0.7	0.3	0.3	0.4	0.4	0.6	0.6
	PrecStd	3.5	0.6	0.4	0.9	0.8	0.8	1.3	1.4	1.3	1.1	0.9	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.9	47.1	71.0	81.9	85.3	89.0	88.5	76.5	63.2	47.1	36.0
		MCWB	32.1	31.8	40.5	56.1	61.7	68.1	69.9	69.2	65.2	53.7	45.1	35.1
	2%	DB	31.1	32.2	40.2	63.3	76.4	82.0	85.4	81.3	70.0	57.1	41.4	33.1
		MCWB	30.2	30.7	35.0	52.1	60.9	66.4	69.1	67.8	60.3	50.1	40.0	32.0
	5%	DB	28.5	30.4	36.8	57.6	72.6	78.7	82.7	77.2	65.7	52.4	37.4	31.3
		MCWB	27.4	29.2	32.7	48.3	58.5	65.0	68.3	65.6	58.2	47.8	36.1	30.3
	10%	DB	25.5	27.5	34.2	52.6	68.4	75.1	79.3	73.1	61.7	48.5	34.1	28.3
		MCWB	24.5	26.2	31.3	44.6	56.5	63.2	67.0	64.0	55.7	45.5	33.0	27.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.2	32.6	41.1	57.9	65.9	72.2	74.4	72.1	65.7	54.8	45.1	35.1
		MCDB	32.6	33.3	46.2	67.9	77.1	81.5	84.3	82.6	74.6	61.5	46.3	35.5
	2%	WB	30.3	31.0	35.9	52.8	62.2	69.1	72.0	69.2	61.9	50.9	40.3	32.1
		MCDB	31.1	32.0	39.1	62.4	73.5	78.9	82.0	79.8	69.0	56.0	41.5	32.8
	5%	WB	27.5	29.1	33.6	49.0	59.9	66.5	69.9	66.7	58.8	48.1	36.3	30.3
		MCDB	28.5	30.4	35.7	57.2	71.0	75.8	79.8	75.5	64.5	51.4	37.3	31.3
	10%	WB	24.6	26.2	31.6	45.3	57.4	64.2	67.8	64.4	56.3	45.7	33.2	27.3
		MCDB	25.6	27.5	33.7	51.7	66.5	72.6	77.0	71.7	60.8	48.3	33.9	28.3
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.2	12.1	14.7	17.5	16.5	16.6	14.9	12.7	8.6	7.4	9.2
		MCDBR	10.0	9.8	14.4	21.4	22.9	20.5	21.0	19.8	17.7	14.4	8.1	8.0
		MCWBR	9.7	9.0	10.8	14.1	13.0	11.4	11.3	10.3	10.6	10.1	7.8	8.1
	5% WB	MCDBR	9.8	9.3	11.4	20.1	20.8	19.4	19.6	18.5	16.3	13.2	7.5	7.9
MCWBR		9.7	8.7	9.5	14.4	13.3	12.3	12.5	11.2	10.9	10.0	7.5	7.9	
Clear-Sky Solar Irradiance	taub	0.281	0.310	0.361	0.389	0.381	0.377	0.397	0.411	0.367	0.338	0.299	0.274	
	taud	2.014	2.031	1.960	2.101	2.329	2.386	2.335	2.278	2.370	2.303	2.124	1.983	
	Ebn at Noon	164	219	239	257	270	273	263	249	244	213	159	125	
	Edh at Noon	18	29	41	42	36	35	36	36	28	22	16	14	
All-Sky Solar Radiation	RadAvg	166	443	862	1278	1623	1737	1740	1301	762	332	134	89	
	RadStd	25	75	104	164	180	155	183	117	90	46	25	18	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page